FINAL-YEAR PROJECT PROPOSAL

# CampusFix

AI-assisted fault reporting for UWE students, in halls and across campus


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THE IDEA IN ONE PARAGRAPH

CampusFix is a mobile app that lets UWE students report any problem, in their flat in halls or anywhere on campus, in under a minute. The student signs in with their student number, so the app already knows their flat: they just say what is wrong and add a photo if they want. For campus faults they scan a QR code on the room door. AI reads the description and photo, works out what the problem is, predicts its urgency using UWE's own repair levels (Emergency, Urgent, Routine) and sends it to the right team. Staff manage reports on a web dashboard, and the student follows the report until it is fixed. If nothing needs repairing, such as lost work on a library computer, the AI gives step-by-step help instead.

**1 app**
for halls and campus faults

**3 parts**
student app, AI service, staff dashboard

**3 levels**
UWE's Emergency, Urgent and Routine priorities

**~1,600**
labelled descriptions and photos to collect

## 01 The problem

Students in UWE halls can already report repairs online, but students have no online route for campus buildings, and no existing route is a photo-first mobile app with automatic triage.

| Who | Area | Current route | Gap |
|---|---|---|---|
| Students in UWE halls | Accommodation | Online student repairs portal with a job tracker | Halls only; no photo upload or app mentioned |
| Staff | Campus buildings | Facilities Online work requests | Staff login required |

| Who | Area | Current route | Gap |
|---|---|---|---|
| Students | Campus buildings | General Estates contact page, switchboard or Facilities phone line | **No mobile, photo-based or trackable route** |

In April 2023 a student proposed on the Students' Union ideas board that the UWE app should let students report broken plug sockets by building and room, citing frequent faults at Frenchay. The idea received no comments and shows no recorded outcome.

CampusFix covers both areas in one app. For campus buildings it fills the gap outright. For halls it improves on the existing portal with a pre-filled location, photos, AI triage against UWE's published repair priorities and duplicate detection; the existing portal becomes the baseline for evaluation. It would not replace Facilities Online: it is a student-facing front end that could feed the existing staff workflow.

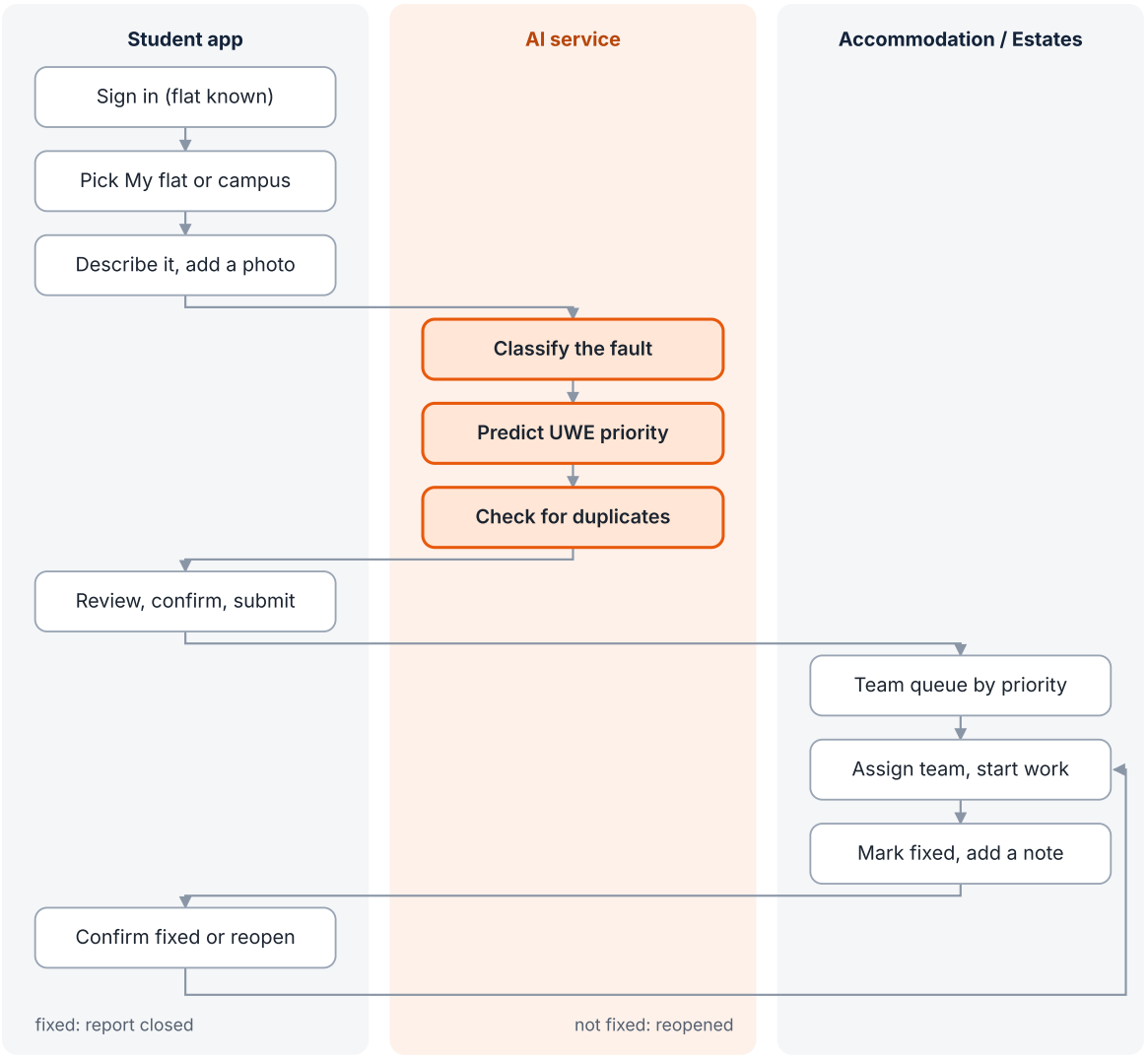
UWE publishes no usage figures for the portal. A Freedom of Information request and a short student survey are planned to measure current reporting.

## 02 How it works

A report moves through three parts: the student app captures it, the AI service triages it in seconds, and staff handle it until the student confirms the fix.

**EXAMPLE**

A student types *"the oven in our kitchen won't turn on"* with no photo. The flat is already known from their profile, the AI suggests **Kitchen appliances** and **Urgent**, and the report goes straight to Accommodation Services.




Report process across the three parts of the system. Orange steps are the AI.

## Four branches inside the main flow

- **Somewhere on campus:** the student scans the room QR code or picks the building and room, and the report goes to the matching Estates team.
- **Emergency:** if the AI or the safety rules suggest Emergency, the app shows the UWE emergency number and asks the student to call instead of waiting in the queue.
- **Duplicate or unsure:** a likely duplicate offers "Add to existing report"; a low-confidence result enters the queue flagged "Needs review".
- **Nothing to fix:** if no repair is needed (e.g. "I lost my work on a library computer"), the AI shows step-by-step help from approved UWE pages or points to the right service, such as the IT Service Desk. No ticket is created unless the student taps "This didn't help, report it".

## 03 Features and MVP scope

The MVP is one complete loop: a student reports a fault, AI suggests a category and priority, and staff see and update it on a dashboard. Everything else is optional and can be cut without breaking that loop.

| Part | MVP (must have) | Extra / Stretch (if time allows) |
|---|---|---|
| **Student app**<br>Flutter | • Sign in with student number; profile holds hall, flat and room (sample data)<br>• Choose "My flat" (pre-filled) or "Somewhere on campus"<br>• Campus location by room QR code, or building, floor and room picker<br>• Describe the problem; add a photo (optional)<br>• Review screen to confirm or correct the AI's category, priority and location<br>• Emergency results show the UWE emergency number instead of queueing<br>• Submit with a reference number; My reports list with status timeline and staff notes | • Help answers when nothing needs fixing, with "This didn't help, report it"<br>• Confirm a fix, or reopen<br>• "Already reported" prompt and "Me too"<br>• Push notifications<br>• Up to 3 photos per report |
| **AI service**<br>Python | • Classify the fault from the description, the photo, or both, with a confidence score<br>• Predict UWE's repair priority, with safety keyword rules that force Emergency<br>• Route halls reports to Accommodation Services and campus reports to the right Estates team<br>• Flag low-confidence results as "Needs review"<br>• Log every correction as a new label | • Spot problems with nothing to repair and answer only from approved UWE help pages, citing them<br>• Point to the right service (IT Service Desk, Library)<br>• Duplicate detection<br>• Rewrite descriptions into clear ticket text<br>• *Stretch:* automatic face blurring |
| **Staff dashboard**<br>React | • Staff sign in<br>• Queue sorted by priority then age; filter by halls or campus, team, building, status<br>• Ticket detail: description, photos, AI suggestions and confidence, location<br>• Time left against UWE's response target (4 h, 8 h, 7 days)<br>• Correct category or priority; assign a team; update status; add a note the student sees | • Merged duplicates with a report count<br>• Hotspots by hall and building; average fix time<br>• Help-answer stats<br>• Manager and team-member roles<br>• *Stretch:* repeat-fault alerts |
| **Admin set-up** | • Halls, flats, buildings, rooms, teams and sample students seeded from a config file<br>• Printable QR code for each campus room | • Knowledge base of approved UWE help articles |

**Out of scope:** integration with UWE's live Facilities Online system, real deployment across campus, and emergency reporting (the app directs users to the UWE emergency number).

### Report statuses

| Status | Set by | Meaning |
|---|---|---|
| Submitted | Student | Report sent; AI suggestions attached |
| Needs review | AI service | Low confidence; staff check the category first |
| Merged | AI service or staff | Joined to an existing open report |
| Assigned / In progress | Staff | A team owns it / work has started |
| Fixed | Staff | Work done, with a note to the student |
| Closed / Reopened | Student (or automatic) | Fix confirmed / not fixed, back to the team |
| Answered | AI service | No repair needed; help given in the app, not sent to staff |

## 04 AI and dataset requirements

The AI needs two datasets: about **1,000 labelled fault descriptions**, the main input because many faults (a dead oven, no hot water) look normal in a photo, and about **600 labelled photos**. Each item gets a category and one of UWE's three published priorities. Both targets are working estimates to confirm once collection starts.

| Category | Example faults | Where | Typical UWE priority |
|---|---|---|---|
| Plumbing and water | Leaks, blocked toilet or sink, flooding | Both | Routine to Emergency (serious flooding) |
| Heating and hot water | No heating, no hot water, cold radiator | Both | Urgent |
| Electrical and lighting | Dead sockets, broken lights, exposed wiring, power loss | Both | Routine to Emergency (power loss, hazards) |
| Kitchen appliances | Oven, hob, fridge or freezer not working | Halls | Urgent |
| Doors, windows and locks | Faulty lock, broken window, door not closing | Both | Urgent (locks) or Routine |
| Furniture and fittings | Broken chair, desk, wardrobe, blind | Both | Routine |
| IT and AV | Projector, screen, lecture-room PC | Campus | Routine |
| Cleaning and pests | Spills, overflowing bins, pests | Both | Routine |
| Safety | Blocked fire exit, wedged fire door, trip hazard | Both | Emergency |

| Category | Example faults | Where | Typical UWE priority |
|---|---|---|---|
| No repair needed | Lost files on a library PC, can't log in, Wi-Fi help | Both | Answered with help |

UWE's repairs service standards: Emergency = attend within 4 hours, Urgent = within 8 hours, Routine = within 7 days.

### Collection and labelling plan

1. **Descriptions:** seed examples from UWE's published examples, real-style wording from a short student survey, and LLM-generated variations that I check and label by hand.
2. **Photos:** faults photographed around campus (and in halls with permission), with no people in shot, location logged for each.
3. **Labels:** category and priority for every item in a CSV (id, text or filename, category, priority, area).
4. **Top-up:** openly licensed public datasets for thin photo categories, licences checked for research use.
5. **Augmentation:** rotation, crop and brightness to increase photo variety.
6. **Split:** 70 / 15 / 15 train, validation and test, keeping items about the same fault in one set.
7. **Duplicates:** a separate set of report pairs (same fault vs different fault).

**EVALUATION**

Accuracy, precision, recall and a confusion matrix per category and per priority, comparing text-only, photo-only and combined models, and my fine-tuned models against a general AI API on the same test set. **Emergency recall is the key safety metric:** a missed emergency matters more than a false alarm. Usability is measured with the SUS questionnaire and task times, with the existing portal as the baseline.

## 05 Feasibility and risks

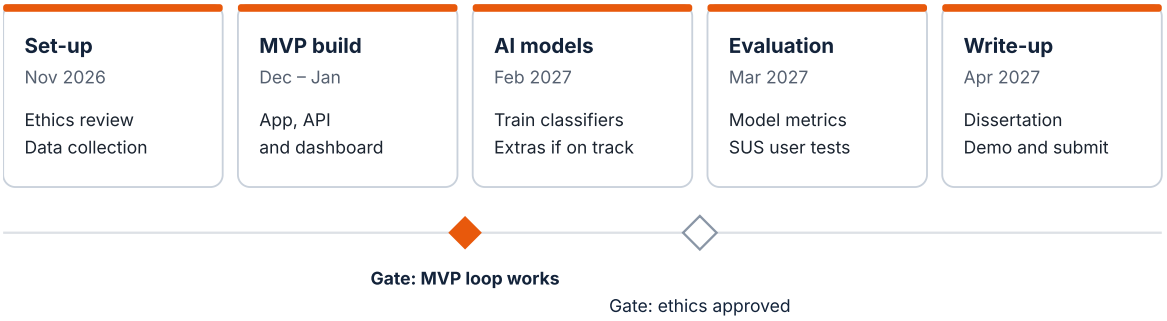
The MVP is feasible in one academic year: a standard mobile app, a web dashboard and an AI service, built mostly in technologies I already use.

| Layer | Technology | Prior experience |
|---|---|---|
| Student app | Flutter (Android and iOS) | Yes |
| Staff dashboard | React | Yes |
| API and database | .NET / C# with PostgreSQL | Yes |
| AI service | Python; fine-tuned text classifier and small image model (e.g. MobileNet or EfficientNet) | Partly (IBM AI Engineering coursework) |
| Comparison model | General AI API (e.g. OpenAI or Google) | New, low effort |
| Storage | Firebase Storage or S3 | Yes |

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| Too little labelled data | Medium | High | Start collecting in November; public datasets; augmentation; compare against an AI API so results are meaningful even if accuracy is modest |
| Ethics approval delays user testing | Medium | Medium | Agree risk level early; build and test the AI first, which needs no participants |
| AI misses an emergency | Low | High | Safety keyword rules force Emergency; emergency number always visible in the app |
| Help answer gives wrong advice | Medium | Medium | Answer only from approved UWE pages and cite them; say "I don't know" otherwise; always offer "report it" |
| Estates not available as a stakeholder | Medium | Low | Evaluate the staff side with demo walkthroughs and peer testers |
| Scope creep from extras | Medium | Medium | Extras start only after the MVP loop works end to end |

## 06 Plan and next steps

Recommendation: proceed with CampusFix in place of the meal planner, with the MVP loop working by the end of January.




Proposed phases, November 2026 to April 2027. Dates to be aligned with the confirmed submission deadline.

| Next step | By |
|---|---|
| Agree the risk level and complete the Student Ethical Review Record (with David) | 6 Nov 2026 |
| Confirm the dissertation submission deadline and milestone dates | 6 Nov 2026 |
| Email UWE Estates about current reporting and a short interview | 6 Nov 2026 |
| Send a Freedom of Information request on repair volumes and response times | 13 Nov 2026 |
| Draft the participant information sheet, consent form and student survey | 20 Nov 2026 |
| Set up the repository, Flutter app and API skeleton | 27 Nov 2026 |
| Collect the first 300 labelled descriptions and photos | 30 Nov 2026 |

## 07 Ethics

I expect CampusFix to be **low risk**, so the supervisor would complete UWE's Student Ethical Review Record and no full committee application would be needed. Only high-risk undergraduate projects need a full application, which can take four weeks or more.

| Question | Draft answer |
|---|---|
| Aim | Build and evaluate an AI-assisted fault-reporting app for UWE halls and campus; evaluate classification accuracy and usability. |
| Participants | 8 to 12 adult UWE students for usability testing and a short survey; optionally 1 to 2 Estates staff for a feedback interview. No under-18s or vulnerable groups. |
| Recruitment | Voluntary, through course peers and an open message; no incentives; no pressure from staff. |
| Activities | 20 to 30 minute session: set report tasks on a test build, SUS questionnaire, short feedback questions. No sensitive topics. |
| Data collected | SUS scores, task times, anonymous written feedback and fault descriptions. Names and emails only on consent forms, stored separately. |
| Image dataset | Photos of fixtures and fittings with no people in shot; any image showing a person, face, badge or screen content is deleted or blurred. |
| Personal data in the app | Test accounts and sample students only; no real student or accommodation records. |
| Third-party AI | Only data with no personal information is sent to an external AI API, and its data terms are checked first. |
| Storage and retention | UWE OneDrive, password protected; participant data deleted after the project is marked. |
| Consent and withdrawal | Participant information sheet and signed consent; withdrawal any time during the session and up to an agreed date afterwards. |
| Risks to participants | Minimal: equivalent to everyday app use. |
| Permissions | Ask Estates before placing QR codes on doors; otherwise use printed demo codes. |

Documents to prepare with UWE templates: participant information sheet, consent form, SUS questionnaire and feedback questions, student survey.

## 08 Questions for discussion

1. Do you agree to proceed with CampusFix in place of the meal planner, with the MVP as defined in section 03?
2. Do you agree the project is low risk for ethics, and does an Estates interview change that?
3. What is the confirmed submission deadline, so the plan dates can be fixed?
4. Should the help assistant for "nothing to fix" problems stay an extra, or is it central enough to be part of the MVP?
5. Should anonymous reporting be allowed, and do staff need separate roles in the MVP?

### Sources


- UWE Bristol, Repairs request (residents' guide): uwe.ac.uk/life/accommodation/current-students/residents-guide/repairs-request
- UWE Bristol, Repairs service standards: uwe.ac.uk/life/accommodation/current-students/residents-guide/service-standards/repairs
- UWE Bristol, Estates and Facilities: uwe.ac.uk/about/services/estates-and-facilities
- UWE Students' Union, What to do with non-academic feedback: thestudentsunion.co.uk/support/studentreps/imastudentrep/whattodowithnonacademicfeedback
- UWE Students' Union, Student idea #492 (April 2023): thestudentsunion.co.uk/representation/student-ideas/492
- UWE Bristol, Applying for ethical approval: uwe.ac.uk/research/policies-and-standards/research-ethics/applying-for-ethical-approval


Prepared with the help of an AI assistant (Claude) for research and drafting; scope and decisions are my own.